

## ESEMPI

$$\bullet \lim_{x \rightarrow +\infty} x \left( \left(1 + \frac{1}{x}\right)^x - e \right) = ?$$

Per  $x \rightarrow +\infty$ ,  $t: \frac{1}{x} \rightarrow 0^+$  e  $x \left( \left(1 + \frac{1}{x}\right)^x - e \right) = \frac{(1+t)^{\frac{1}{t}} - e}{t} = \frac{1}{t} \left( \exp\left(\frac{\log(1+t)}{t}\right) - e \right) = \frac{1}{t} \left( \exp\left(1 - \frac{t}{2} + o(t)\right) - e \right) =$

$$= \frac{e}{t} \left( \exp\left(-\frac{t}{2} + o(t) - 1\right) \right) = \frac{e}{t} \left( 1 - \frac{t}{2} + o(t) - 1 \right) = e \left( -\frac{1}{2} + o(1) \right) \rightarrow -\frac{e}{2}$$

## ALGEBRA DELL'O - PICCOLO

Per il simbolo dell'o-piccolo valgono le seguenti regole che derivano dalle proprietà dei limiti.

Per  $x \rightarrow 0$

$$\textcircled{1} \quad \forall a \geq 0 \text{ e } \forall c \neq 0 \quad c \cdot o(x^a) = o(c \cdot x^a) = o(x^a)$$

$$\textcircled{2} \quad \forall b > a \geq 0 \text{ e } \forall c \quad c \cdot x^b + o(x^a) = o(x^a)$$

$$\textcircled{3} \quad \forall b \geq a \geq 0 \quad o(x^b) + o(x^a) = o(x^a)$$

$$\textcircled{4} \quad \forall a \geq -b \quad x^b o(x^a) = o(x^{a+b})$$

$$\textcircled{5} \quad \forall a, b \geq 0 \quad o(x^a) \cdot o(x^b) = o(x^{a+b})$$

$$\textcircled{6} \quad \forall a, b \geq 0 \text{ e } \forall c \quad (c \cdot x^a + o(x^a))^b = c^b \cdot x^{ab} + o(x^{ab})$$

$$\textcircled{7} \quad \forall a \geq 0 \text{ e } \forall c \quad o(c \cdot x^a + o(x^a)) = o(x^a)$$

Per  $x \rightarrow x_0$  basta sostituire  $o(x^a)$  con  $o((x-x_0)^a)$

## ESEMPI

$$\bullet \lim_{x \rightarrow 0} \left( \frac{\exp(x)}{x} \right)^{\frac{1}{x^2}} = ?$$

Per  $x \rightarrow 0$

$$\left( \frac{\exp(x)}{x} \right)^{\frac{1}{x^2}} = \exp\left(\frac{1}{x^2} \log\left(\frac{\exp(x)}{x}\right)\right) = \exp\left(\frac{1}{x^2} \log\left(1 - \frac{x^2}{6} + o(x^2)\right)\right) = \exp\left(\frac{1}{x^2} \left(-\frac{x^2}{6} + o(x^2) + o\left(-\frac{x^2}{6} + o(x^2)\right)\right)\right) =$$

$$\exp\left(-\frac{1}{6} + o(1)\right) = e^{-1/6}$$

$$\bullet \lim_{x \rightarrow 0} \frac{\exp(x^2) - \exp(x)}{x^2 (\cos(x^2) - \cos(x))} = ?$$

$$\exp(x) = x - \frac{x^3}{6} + o(x^3)$$

$$\cos(x) = 1 - \frac{x^2}{2} + o(x^2)$$

$$= \frac{x^2 - \frac{x^6}{6} + o(x^6) - \left(x - \frac{x^3}{6} + o(x^3)\right)^2}{x^2 \left(1 - \frac{x^4}{2} + o(x^4) - \left(1 - \frac{x^2}{2} + o(x^2)\right)\right)} = \frac{\cancel{x^2} - \frac{x^6}{6} + o(x^6) - \cancel{x^2} - 2\frac{x^4}{6} + o(x^4)}{x^2 \left(\cancel{1} - \frac{x^4}{2} + o(x^4) - \left(\cancel{1} - 2\frac{x^2}{2} + o(x^2)\right)\right)} =$$

$$\frac{-\frac{x^6}{6} + o(x^6) + \frac{x^4}{3} + o(x^4)}{x^2 \left(-\frac{x^4}{2} + o(x^4) + x^2 + o(x^2)\right)} = \frac{1}{3}$$